



Reproducibility Is the Standard

WisdomForge Booklet — Ages 15–18

Reproducibility Is the Standard

WisdomForge Booklet — Ages 15–18

About This Book

The path is the mathematics. Hunting the gap. Hostile reviewer, not solver. Blank board. Proof as a social act. The full path in one sitting.

*“What is the unknown? What are the data? What is the condition?” — George Pólya,
How to Solve It (Princeton, 1945)*

Chapter 1: The Path Is the Mathematics



Illustration for Chapter 1

The answer is the end of the path. The path is the mathematics. A model that gives you the answer without the path has given you a shortcut, not a proof.

Objective. The student can distinguish between an answer and a proof, explain why the path is the mathematics, and use a model as a path-checker, not an answer-giver.

For the grown-up in the room. This sitting teaches the student that the path is the mathematics, not the answer. The model is excellent at answers. It is less reliable at paths. The student needs to learn to demand the path, not the answer. If the model gives an answer, the student asks: 'Show me the steps.' If the model skips a step, the student asks: 'Why is that step legal?' The model is a shortcut. The student's job is to walk the path and use the model as a checker, not a replacement.

Why the path matters. The answer to a math problem is a destination. The path is how you get there. If someone gives you the destination, you know where it is. You do not know how to get there. If you walk the path, you know both. The next problem is a different destination, but the path uses the same streets. The student who walks paths learns the streets. The student who collects answers learns nothing reusable. The model gives destinations. It is a taxi. Taxis are useful. They do not teach you to walk. Walk the path yourself. Use the taxi when you already know the streets.

Using the model as a path-checker. The model can check your path. You walk the steps. You show the model your steps. You ask: 'Is each step legal? Did I skip one?' The model reviews the path. It does not replace the walking. If the model says 'your step 3 is wrong,' you fix it. If the model says 'this is correct,' you still verify. The model is a checker, not a walker. The student who uses it this way gets the best of both: their own understanding and a tireless reviewer. The student who lets the model walk the path gets neither.

The household path. Household math is full of paths. A budget: 'We have X. We spend Y on rent, Z on food, W on utilities. What is left?' The answer is a number. The path is the steps. If you show the steps out loud, the child learns that math is a series of visible, walkable steps, not a magic number from a phone. If you ask the model for the answer and skip the steps, the child learns that math is a black box. The path is the inheritance. The answer is not.

What the child copies. The child copies your relationship to math. If you show steps, they show steps. If you ask the model for the answer, they ask the model for the answer. The model is the smoothest answer-giver in history. If the child watches you use it as a replacement for the path, they will use it that way for life. If they watch you use it as a checker after you walked the path, they will use it that way. The modeling is the lesson. The path is the skill. The model is the tool.

Sitting big idea. The path is the mathematics. The answer is the destination. Walk the path. Use the model to check, not to walk.

If they say “The model's path is clearer than mine.”: It is. That is because it has walked a million paths. Clarity is not understanding. Walk your own path, even if it is messier. The mess is where you learn. The model's clean path is a demonstration, not your learning.

If they say “I only need the answer for the test.”: For the test, the answer might be enough. For the next course, the next problem, the next job, it is not. The path is reusable. The answer is disposable. Choose the reusable one.

If they say “I just use the calculator.”: The calculator gives the answer. It does not show the path. If you only use the calculator, you have answers and no paths. Show the child the path once. The answer is disposable. The path is reusable.

If they say “The child won't care about the steps.”: They will not care about the steps as entertainment. They will absorb the idea that math is steps, not magic. That absorption is the lesson. They do not need to care. They need to see it.

Integrity. Do not submit the model's path as your proof. If you walked it, it is yours. If the model walked it and you copied, it is plagiarism. The path is the proof.

Dinner. What path did we walk this week, and did the model check it or replace it?

- Academic integrity: a model answer submitted as a proof is plagiarism of the path. The answer is not the proof.

- Do not ban the model. The point is to use it as a path-checker, not to reject it.

Young sitting:

A proof is not a fancy word. It is a path other people can walk, step by step, from what they know to what you are saying.

Objective. The student can explain what a proof is in plain language: a path of steps from something known to a conclusion, where each step is visible and walkable.

For the grown-up in the room. A proof is a path. That is the whole idea. You start somewhere everyone agrees on. You take a step. You show the step. You take another step. You show it. You arrive at the conclusion. If every step is visible and anyone could walk it, the proof holds. If a step is missing or invisible, the proof is broken. The model can produce the conclusion. It cannot always show the path. The student needs to learn that the path is the proof, not the answer.

The path. Imagine you tell a friend your house is two blocks from school. They say: prove it. You do not just say 'it is.' You walk the path. Left on Oak. Right on Maple. Two blocks. Here is the door. That is a proof. You showed each step. Your friend could walk it. A

mathematical proof is the same thing with ideas instead of streets. You start somewhere everyone agrees. You take a step. You show it. You take another. You show it. You arrive. The path is the proof. The answer is the door. Without the path, the answer is just a claim.

The model's shortcut. A model will give you the answer. 'The sum is 42.' It will not always show the path. It jumps to the door. That is useful if you only need the answer. It is not a proof. A proof is the path. If you cannot walk the path yourself, you have the answer but not the proof. The model is a shortcut machine. Shortcuts are useful. They are not mathematics. Mathematics is the path. The student who confuses the shortcut for the path will have answers and no understanding. The student who knows the difference can use the shortcut when the path does not matter and walk the path when it does.

Why the path matters. The answer to a math problem is a destination. The path is how you get there. If someone gives you the destination, you know where it is. You do not know how to get there. If you walk the path, you know both. The next problem is a different destination, but the path uses the same streets. The student who walks paths learns the streets. The student who collects answers learns nothing reusable. The model gives destinations. It is a taxi. Taxis are useful. They do not teach you to walk. Walk the path yourself. Use the taxi when you already know the streets.

Using the model as a path-checker. The model can check your path. You walk the steps. You show the model your steps. You ask: 'Is each step legal? Did I skip one?' The model reviews the path. It does not replace the walking. If the model says 'your step 3 is wrong,' you fix it. If the model says 'this is correct,' you still verify. The model is a checker, not a walker. The student who uses it this way gets the best of both: their own understanding and a tireless reviewer. The student who lets the model walk the path gets neither.

Sitting big idea. A proof is a path other people can walk. The model gives the door. The proof is the walk.

If they say "The answer is what matters.": The answer matters for the task. The path matters for understanding. If you only have the answer, you cannot do the next problem. If you have the path, you can walk it again.

If they say "The model's path is clearer than mine.": It is. That is because it has walked a million paths. Clarity is not understanding. Walk your own path, even if it is messier. The mess is where you learn. The model's clean path is a demonstration, not your learning.

If they say "I only need the answer for the test.": For the test, the answer might be enough. For the next course, the next problem, the next job, it is not. The path is reusable. The answer is disposable. Choose the reusable one.

Integrity. Do not copy the model's answer and call it your proof. The proof is the path. If you did not walk it, it is not yours.

Dinner. What path did we walk today, and could someone else follow it?

- Do not use the word 'theorem' yet. Use 'path.' The vocabulary comes later.
- Keep it to simple examples: arithmetic steps, a geometric argument, a logic puzzle.

Adult sitting:

If you only show the answer, the child learns that answers are the point. They are not. The path is.

Objective. The adult models walking a path on one real problem (household math, a budget calculation, a measurement), shows each step out loud, and names that the path is the skill.

For the grown-up in the room. The adult models the path. Pick a real calculation: a budget, a measurement conversion, a time estimate. Do it out loud, step by step. Show each step. Say: 'The answer is not the point. The steps are. If I skip the steps, I have an answer I cannot check.' The child who watches you walk the path learns that understanding is the steps, not the destination. The child who watches you ask the model for the answer learns that answers are the point. They are not.

The household path. Household math is full of paths. A budget: 'We have X. We spend Y on rent, Z on food, W on utilities. What is left?' The answer is a number. The path is the steps. If you show the steps out loud, the child learns that math is a series of visible, walkable steps, not a magic number from a phone. If you ask the model for the answer and skip the steps, the child learns that math is a black box. The path is the inheritance. The answer is not.

What the child copies. The child copies your relationship to math. If you show steps, they show steps. If you ask the model for the answer, they ask the model for the answer. The model is the smoothest answer-giver in history. If the child watches you use it as a replacement for the path, they will use it that way for life. If they watch you use it as a checker after you walked the path, they will use it that way. The modeling is the lesson. The path is the skill. The model is the tool.

Sitting big idea. Model the path, not the answer. The child who watches you walk steps learns that understanding is the walk, not the destination.

If they say "I just use the calculator.": The calculator gives the answer. It does not show the path. If you only use the calculator, you have answers and no paths. Show the child the path once. The answer is disposable. The path is reusable.

If they say “The child won't care about the steps.”: They will not care about the steps as entertainment. They will absorb the idea that math is steps, not magic. That absorption is the lesson. They do not need to care. They need to see it.

Integrity. If you use the model for the answer, say so. Do not pretend the answer was yours if the model walked the path. The child can tell, and the lie teaches the wrong lesson.

Dinner. What path did we walk out loud this week, and what did the steps teach us?

- Do not perform the path as a lecture. Walk it naturally. The child watches.

- If you use the model for the answer, say: 'I am checking the answer. I already walked the path.' The model is the checker, not the walker.

The Big Idea

The path is the mathematics. The answer is the destination. Walk the path. Use the model to check, not to walk. Reproducibility is not optional because the instrument is fluent.

Practice

1. One problem at the edge of skill. Path on paper. No tool. 2. Hunt a gap. Mark the missing step. 3. Tool as hostile reviewer of legal steps only. Reject a full solution. 4. Close. Blank-board one key step.

Reflect

If the chat closed, which step could you still walk — and which was a rental?

Chapter 2: Hunting the Gap



Illustration for Chapter 2

The model is a gap-jumper. It produces answers with invisible holes. The skill is to hunt the gap before you trust the path.

Objective. The student can audit a model's mathematical solution for missing steps, identify each gap, and either fill it independently or flag it as unsupported.

For the grown-up in the room. This sitting teaches the student to audit the model's solutions the way an editor audits a draft. The model will produce a solution with steps. Some steps will be real. Some will be jumps disguised as steps. The student's job is to hunt the gaps. For each step, ask: 'Does this follow from the previous one? Can I reproduce it on a blank board?' If yes, the step is real. If no, the step is a gap. Fill it yourself or flag it. The model will not flag its own gaps. The student who can hunt gaps will not be fooled by a fluent solution that has holes.

The gap audit. A gap audit is simple. Read each step of the model's solution. For each step, ask two questions: 'Does this follow from the previous step?' and 'Can I reproduce this step on a blank board?' If both answers are yes, the step is real. If either is no, there is a gap. Write GAP. Then decide: can I fill it? If you can fill it with your own work, the path holds, and you have made the step yours. If you cannot fill it, the path is broken at that point. Do not trust the solution past the gap. The gap is not a minor inconvenience. It is a structural failure in the proof.

The invisible gap. The most dangerous gap is the one that looks like a step. The model writes 'therefore' or 'so' and jumps. The jump looks like a step because it has the right words around it. But the words 'therefore' and 'so' do not create a logical connection. They announce one. If the connection is not shown, the step is a gap dressed as a step. The student who reads 'therefore' and trusts it has been fooled by the costume. The student who reads 'therefore' and asks 'what follows from what?' has started the audit. The audit is the skill. The model will not perform it for you.

Steps at the kitchen table. The kitchen table is where the child learns whether steps are real or just for school. If you calculate the tip, the budget, or the time and show each step, the child learns that steps are how adults think. If you ask the model for the answer and skip the steps, the child learns that answers are how adults think. The model is the smoothest answer-giver in history. If the child watches you use it as a replacement for thinking, they will use it that way for life. If they watch you walk the path and use the model as a checker, they will use it that way. The kitchen table is the classroom that matters.

The inheritance of steps. The child inherits the adult's relationship to steps. If the adult shows steps, the child shows steps. If the adult skips, the child skips. The model makes skipping easier than ever. The adult who resists the shortcut once a week, at the table, teaches the child that the path is the skill. The adult who always takes the shortcut teaches the child that the shortcut is the skill. The inheritance is not a lecture. It is a habit, modeled.

Sitting big idea. The model jumps and dresses the jump as a step. Hunt the gap. Fill it or flag it. A solution with unfilled gaps is not a proof.

If they say “The model's solution looked fine to me.”: It looked fine because the gaps were dressed as steps. The audit is how you see the costume. If you did not find gaps, either the path was complete or you missed them. Audit again with a harder problem.

If they say “I can't fill the gap, but the answer is probably right.”: 'Probably right' is not a proof. If you cannot fill the gap, you have an unsupported step. The answer might be right. The proof is broken. Those are different things. A right answer with a broken proof is a coincidence, not a proof.

Integrity. Cite the steps you verified. Flag the gaps you could not fill. Do not submit a model's solution as your proof if it has unsupported gaps.

Dinner. What gaps did we find in a model's solution this week, and did we fill them or flag them?

- Academic integrity: a model solution with gaps you cannot fill is not your proof. You can cite the model's steps you verified. You cannot cite the gaps.

- Do not let the student think the model is malicious. The gaps are structural, not intentional. The model does not know it is jumping.

Young sitting:

A proof with a hole in it is not a proof. It is a guess with a gap.

Objective. The student can identify a missing step in a simple mathematical argument and explain why the gap breaks the path.

For the grown-up in the room. A proof is a path. If a step is missing, the path has a hole. You cannot walk across a hole. The model is very good at jumping over holes. It produces the answer without showing the step that connects. The student needs to learn to look for holes. If the model says 'and then the answer is 42,' the student asks: 'What step got you from the line before to 42?' If there is no step, there is a hole. The hole is the lesson.

The hole in the path. A path has steps. If you remove a step, the path has a hole. You can see the other side, but you cannot walk there. The model is very good at jumping. It says 'the answer is 42' and skips the step that gets you there. If you cannot see the hole, you will trust the jump. If you can see it, you will ask: 'Where did 42 come from?' That question is the skill. The model's jump is not a proof. It is a leap. A proof is a walk, not a leap.

Finding the hole. To find the hole, read each step and ask: 'Does this step follow from the one before it?' If yes, walk on. If you cannot see how it follows, there is a hole. The hole might be a missing calculation, a skipped assumption, or a logical jump. Mark the hole. Ask: 'What step would fill this gap?' If you can fill it, the path holds. If you cannot, the path is broken. The model will not mark its own holes. You have to find them. That is the student's job, not the model's.

The gap audit. A gap audit is simple. Read each step of the model's solution. For each step, ask two questions: 'Does this follow from the previous step?' and 'Can I reproduce this step on a blank board?' If both answers are yes, the step is real. If either is no, there is a gap. Write GAP. Then decide: can I fill it? If you can fill it with your own work, the path holds, and you have made the step yours. If you cannot fill it, the path is broken at that point. Do not trust the solution past the gap. The gap is not a minor inconvenience. It is a structural failure in the proof.

The invisible gap. The most dangerous gap is the one that looks like a step. The model writes 'therefore' or 'so' and jumps. The jump looks like a step because it has the right words around it. But the words 'therefore' and 'so' do not create a logical connection. They announce one. If the connection is not shown, the step is a gap dressed as a step. The student who reads 'therefore' and trusts it has been fooled by the costume. The student who reads 'therefore' and asks 'what follows from what?' has started the audit. The audit is the skill. The model will not perform it for you.

Sitting big idea. A proof with a missing step is a guess with a gap. The model jumps. The student finds the hole.

If they say "The model probably skipped it because it's obvious.": Obvious to whom? If it is obvious to you, fill it. If it is not obvious to you, it is a hole. 'Obvious' is the most common word used to hide a gap. Check it.

If they say "The model's solution looked fine to me.": It looked fine because the gaps were dressed as steps. The audit is how you see the costume. If you did not find gaps, either the path was complete or you missed them. Audit again with a harder problem.

If they say "I can't fill the gap, but the answer is probably right.": 'Probably right' is not a proof. If you cannot fill the gap, you have an unsupported step. The answer might be right. The proof is broken. Those are different things. A right answer with a broken proof is a coincidence, not a proof.

Integrity. If you find a hole, say so. Do not pretend the path was complete because the answer was right. A right answer with a hole is still a broken proof.

Dinner. What hole did we find in a path this week, and did we fill it?

- Keep the examples simple: arithmetic with a skipped step, a logic puzzle with a jump.
- Do not shame the model for skipping. The point is the student's ability to see the gap.

Adult sitting:

If you skip steps at home, the child learns that steps are for school. They are not. They are for life.

Objective. The adult models showing every step on one real household calculation, names the steps out loud, and demonstrates that the path is the skill, not just the answer.

For the grown-up in the room. This sitting is about the adult showing work at home. Not in a classroom. At the kitchen table, on the budget, on the measurement, on the schedule. The adult who shows steps at home teaches the child that steps are a life skill, not a school requirement. The adult who skips steps at home teaches the child that steps are for school and answers are for life. The model gives answers. The adult who shows steps gives the child something the model cannot: the habit of walking the path.

Steps at the kitchen table. The kitchen table is where the child learns whether steps are real or just for school. If you calculate the tip, the budget, or the time and show each step, the child learns that steps are how adults think. If you ask the model for the answer and skip the steps, the child learns that answers are how adults think. The model is the smoothest answer-giver in history. If the child watches you use it as a replacement for thinking, they will use it that way for life. If they watch you walk the path and use the model as a checker, they will use it that way. The kitchen table is the classroom that matters.

The inheritance of steps. The child inherits the adult's relationship to steps. If the adult shows steps, the child shows steps. If the adult skips, the child skips. The model makes skipping easier than ever. The adult who resists the shortcut once a week, at the table, teaches the child that the path is the skill. The adult who always takes the shortcut teaches the child that the shortcut is the skill. The inheritance is not a lecture. It is a habit, modeled.

Sitting big idea. Show steps at the kitchen table. The child who sees an adult walk the path learns that steps are for life, not just for school.

If they say “Nobody shows work in real life.”: Everybody who makes decisions that matter shows work. The engineer, the doctor, the accountant. The people who skip are the people whose decisions do not matter yet. When they matter, the steps matter. Teach the steps now.

If they say “The model does it faster.”: It does. Faster is the model's advantage. Understanding is yours. If you trade understanding for speed, the child learns that speed is the skill. It is not. Speed is a convenience. Understanding is the skill.

Integrity. If you used the model for the answer, say so. Do not present the model's answer as your own work. The child can tell, and the lie teaches that answers are all that matter.

Dinner. What steps did we show at the table this week, and what did the child see?

- Do not make this a lecture on showing work. Just show it. The child watches.
- If you use the model, use it after the path. Say: 'I walked the path. Let me check the answer.'

The Big Idea

The model jumps and dresses the jump as a step. Hunt the gap. Fill it or flag it. A solution with unfilled gaps is not a proof. Reproducibility is not optional because the instrument is fluent.

Practice

1. One problem at the edge of skill. Path on paper. No tool. 2. Hunt a gap. Mark the missing step. 3. Tool as hostile reviewer of legal steps only. Reject a full solution. 4. Close. Blank-board one key step.

Reflect

If the chat closed, which step could you still walk — and which was a rental?

Chapter 3: The Model Is a Hostile Reviewer



Illustration for Chapter 3

The best use of a model in mathematics is as a hostile reviewer: it attacks your path and finds the weak steps. You fix them. The path gets stronger.

Objective. The student can use a model as a hostile reviewer of their mathematical work, ask it to find the weakest step, and fix the weaknesses independently.

For the grown-up in the room. This sitting teaches the student to use the model as a hostile reviewer. Not a friendly helper. A hostile reviewer. The student walks the path. Then asks the model: 'Attack my proof. Find the weakest step. Tell me where it breaks.' The model finds the weak points. The student fixes them. This is the professional workflow: you write, a reviewer attacks, you fix. The model is a tireless reviewer. The student is the owner of the path. The model does not fix. It finds. The student fixes.

The hostile review. In professional mathematics, proofs are reviewed. The reviewer does not rewrite the proof. They attack it. They find the weak step, the unjustified claim, the missing assumption. The author fixes it. The proof gets stronger. The model can be that reviewer. You walk the path. You show the model each step. You ask: 'Attack this. Where is the weakest step? Where would a skeptic push?' The model finds the weak points. You fix them. The model does not rewrite. You rewrite. The model is the reviewer. You are the author. That is the right relationship.

Why hostile, not friendly. A friendly model says 'your proof looks good.' That is useless. A hostile model says 'step 3 assumes something you did not prove. Step 5 jumps without justification.' That is useful. The student who asks for a friendly review gets nothing. The student who asks for a hostile review gets the weak points. Ask the model to be hostile: 'Find the weakest step. Tell me where a skeptic would push.' The model is better at being hostile than at being right. Use that. The hostile review is the value. The friendly review is flattery.

The reviewer at the table. The child needs to see the model used as a reviewer, not a walker. That means: you do the calculation. You show the steps. Then you say: 'Let me check with the model.' You ask the model to verify, not to solve. If the model solves it anyway, you say: 'I did not ask for the answer. I asked for the check.' That correction teaches the child that the model's role is reviewer, not replacement. The child who sees this once will internalize the relationship. The child who never sees it will use the model as a walker for life.

What the child hears. The child hears: 'I walked the path. The model is checking my walk.' That sentence, spoken once at the table, teaches the relationship. The model is a tool for checking, not a replacement for thinking. The adult who says this out loud gives the child a sentence they will repeat silently for years. The adult who never says it gives the child a model they will use as a walker. The sentence is the inheritance. Say it.

Sitting big idea. Use the model as a hostile reviewer. It attacks. You fix. The path gets stronger. The model does not rewrite. You do.

If they say “The model said my proof was good.”: Ask it to be hostile. 'Find the weakest step.' A model that says 'good' is being friendly, not useful. A model that finds the weak step is being a reviewer. Demand the review, not the compliment.

If they say “I'll just let the model fix it.”: The model finds the weakness. You fix it. If the model fixes it, the fix is the model's, not yours. The proof is only yours if you wrote every fix. The model is the reviewer. You are the author.

Integrity. Cite the model as a reviewer if its hostile review changed your proof. The review is the model's. The fix is yours. The proof is yours. The citation is honest.

Dinner. What proof did we strengthen this week with a hostile review?

- Academic integrity: the model reviews. The student writes. The final proof is the student's, with the model cited as a reviewer if it changed a step.

- Do not let the model rewrite the proof. It finds weaknesses. It does not rewrite. The student rewrites.

Young sitting:

You walk the path. The model checks your steps. That is the right order. Flip it and you lose the path.

Objective. The student can solve a problem on their own, then ask the model to check each step, and explain why walking first and checking second is the right order.

For the grown-up in the room. This sitting establishes the order: walk first, check second. The student solves the problem. Then the model reviews. If the model goes first, the student copies the path and calls it their own. If the student goes first, the model is a checker, not a walker. The order is the lesson. Walk first. Check second. The model is a reviewer, not a replacement.

The right order. You walk the path first. You write each step. You arrive at the answer. Then you ask the model: 'Check my steps.' The model reviews. It finds a gap. You fix it. It confirms a step. You keep it. The model is a checker. You are the walker. If you flip the order, the model walks and you copy. The copy is not a proof. It is a transcript of someone else's walk. The order is the lesson: walk first, check second. The model is a reviewer, not a replacement.

When you are stuck. Sometimes you cannot walk. You are stuck at a step. You may ask the model for a hint: 'What kind of step should come next?' Not: 'What is the answer?' A hint is a nudge. It points to the street. You still walk it. A solution is a taxi. It takes you to the door. You did not walk. The student who asks for hints learns to walk. The student who asks for solutions learns to ride. Walk. Ask for hints when stuck. Never ask for the solution.

The hostile review. In professional mathematics, proofs are reviewed. The reviewer does not rewrite the proof. They attack it. They find the weak step, the unjustified claim, the missing assumption. The author fixes it. The proof gets stronger. The model can be that reviewer. You walk the path. You show the model each step. You ask: 'Attack this. Where is the weakest step? Where would a skeptic push?' The model finds the weak points. You fix them. The model does not rewrite. You rewrite. The model is the reviewer. You are the author. That is the right relationship.

Why hostile, not friendly. A friendly model says 'your proof looks good.' That is useless. A hostile model says 'step 3 assumes something you did not prove. Step 5 jumps without justification.' That is useful. The student who asks for a friendly review gets nothing. The student who asks for a hostile review gets the weak points. Ask the model to be hostile: 'Find the weakest step. Tell me where a skeptic would push.' The model is better at being hostile than at being right. Use that. The hostile review is the value. The friendly review is flattery.

Sitting big idea. Walk first. Check second. The model is a reviewer, not a replacement. When stuck, ask for hints, not answers.

If they say "It's faster to just ask the model.": It is faster. Faster is not the skill. Walking is the skill. The model can check your walk. It cannot walk for you and call it yours. Walk first.

If they say "The model said my proof was good.": Ask it to be hostile. 'Find the weakest step.' A model that says 'good' is being friendly, not useful. A model that finds the weak step is being a reviewer. Demand the review, not the compliment.

If they say "I'll just let the model fix it.": The model finds the weakness. You fix it. If the model fixes it, the fix is the model's, not yours. The proof is only yours if you wrote every fix. The model is the reviewer. You are the author.

Integrity. If the model filled a gap for you, cite it. The gap is the model's step, not yours. The proof is only yours if you walked every step.

Dinner. What path did we walk first this week, and did the model check or replace?

- If the student is stuck, they may ask the model for a hint, not a solution. A hint is a nudge. A solution is a replacement.

- Do not let the student copy the model's path and call it their proof. The walk is the proof.

Adult sitting:

If you only use the model for answers, the child learns that answers are the point. Use it as a reviewer once, out loud. The child learns the right relationship.

Objective. The adult models using the model as a reviewer, not an answer-giver, on one real household calculation, and names the relationship out loud for the child.

For the grown-up in the room. The adult models the reviewer relationship. On one household calculation, the adult walks the path, then asks the model to check. Out loud, the adult says: 'I walked the steps. Now I am asking the model to check, not to solve.' The child hears the distinction. The adult who always asks the model for the answer teaches the child that the model is the walker. The adult who walks and then checks teaches the child that the model is the reviewer. The distinction is the lesson. Say it out loud once.

The reviewer at the table. The child needs to see the model used as a reviewer, not a walker. That means: you do the calculation. You show the steps. Then you say: 'Let me check with the model.' You ask the model to verify, not to solve. If the model solves it anyway, you say: 'I did not ask for the answer. I asked for the check.' That correction teaches the child that the model's role is reviewer, not replacement. The child who sees this once will internalize the relationship. The child who never sees it will use the model as a walker for life.

What the child hears. The child hears: 'I walked the path. The model is checking my walk.' That sentence, spoken once at the table, teaches the relationship. The model is a tool for checking, not a replacement for thinking. The adult who says this out loud gives the child a sentence they will repeat silently for years. The adult who never says it gives the child a model they will use as a walker. The sentence is the inheritance. Say it.

Sitting big idea. Say 'I walked the path. The model is checking my walk.' out loud. The child hears the relationship. That sentence is the inheritance.

If they say "It's easier to just ask the model.": It is easier. Easier is not the skill. The skill is the walk. The model checks the walk. If you skip the walk, the child learns that the model is the walker. Walk. Then check.

Integrity. If you asked the model for the answer and not the check, say so. Do not pretend you walked the path. The child can tell.

Dinner. When did we use the model as a checker instead of a solver this week?

- Do not make this a lecture. Walk the path. Check. Name the relationship. That is enough.
- If the model gives the answer instead of the check, say: 'I asked for a check, not a solution. Let me try again.' The child sees you correct the model. That is also the lesson.

The Big Idea

Use the model as a hostile reviewer. It attacks. You fix. The path gets stronger. The model does not rewrite. You do. Reproducibility is not optional because the instrument is fluent.

Practice

1. One problem at the edge of skill. Path on paper. No tool. 2. Hunt a gap. Mark the missing step. 3. Tool as hostile reviewer of legal steps only. Reject a full solution. 4. Close. Blank-board one key step.

Reflect

If the chat closed, which step could you still walk — and which was a rental?

Chapter 4: Reproducibility Is the Standard



Illustration for Chapter 4

A proof you cannot reproduce is a proof you do not own. The blank board test is the standard for whether the path is yours.

Objective. The student can reproduce a complete proof on a blank board after working with a model, and explain why reproducibility is the standard for mathematical understanding.

For the grown-up in the room. This sitting makes the blank board test the standard. The student works with the model on a proof. Then they close everything and reproduce the full proof on a blank board. If they can, the path is theirs. If they cannot, the path was the model's, and they followed it. The standard is not perfection. It is reproducibility. A mathematician who cannot reproduce their own proof does not own it. The student who adopts this standard will never confuse following for understanding.

The standard. In mathematics, reproducibility is the standard. A proof is yours if you can reproduce it. A proof is the model's if you cannot. The blank board test is the check: close everything, take a blank page, reproduce the proof. If you can, the path is in your hands. If you cannot, the path was in the model's hands and you were reading over its shoulder. Both are useful experiences. Only one is understanding. The student who adopts reproducibility as the standard will always know the difference. The student who does not will confuse following for owning for the rest of their education.

What the test reveals. The blank board test reveals exactly which steps you own and which you followed. You might own steps 1 through 3 and lose the path at step 4. That is useful data. Step 4 is the step you need to relearn. The model can help you relearn it. Then you take the blank board test again. If you can reproduce step 4 this time, you own it now. The test is not a grade. It is a diagnostic. It tells you where the path is yours and where it is not. Then you fix the gaps. The test is the map of your understanding.

The honest test. The adult who takes the blank board test in front of the child and passes teaches: reproducibility is achievable. The adult who takes it and fails, honestly, teaches: reproducibility is the standard, and not meeting it is data, not shame. Both are lessons. The adult who never takes the test teaches: math is something you read, not something you own. That is the wrong lesson. Take the test. Pass or fail. Name the result. The child learns that the standard exists and that honest adults hold themselves to it.

Sitting big idea. Reproducibility is the standard. A proof you cannot reproduce is not yours. The blank board test is the map of what you own and what you followed.

If they say “I don't need to reproduce it. I understood it.”: Understanding while following is real. It is not the same as reproducing. The blank board test is the difference. If you can reproduce it, you own it. If you cannot, you understood it while it was in front of you. That is a different thing. Both are useful. Only one is mastery.

If they say “In the real world, I can just look it up.”: You can look up the steps. You cannot look up the understanding. The steps are on the internet. The path in your hands is not. The blank board test is how you know which one you have.

If they say “I’ll fail it in front of the kid.”: Then you fail honestly. The child who sees an adult fail a blank board test and say 'I need to learn this again' learns more than from a perfect performance. They learn that the standard is real and that honest adults hold themselves to it. The failure is the lesson. The hiding is not.

Integrity. A proof you cannot reproduce is not your proof. Cite the model's help. Do not submit a proof you cannot reproduce on a blank board as your own.

Dinner. What proof did we reproduce on a blank board this week, and where were the gaps?

- Academic integrity: a proof you cannot reproduce is not your proof. You can cite the model's help, but you cannot submit it as your own work.

- Accommodations: a reader or extra time is access, not a ghost. The blank board test is about the student's understanding, not their speed.

Young sitting:

If you close the book and cannot do it again, you did not learn it. You read it.

Objective. The student can reproduce one key step from a problem they solved with the model, without the model or the notes, on a blank board.

For the grown-up in the room. This is the reproducibility test. The student solves a problem with the model's help. Then they close everything. Blank board. Can they reproduce the key step? If yes, they learned it. If no, they read it. The model is a reading machine. Reading is not learning. The blank board test is the check. Use it once on one problem. The child who passes has learned. The child who fails has read. Both are useful data.

Reading is not learning. You can follow a model's steps while it shows them. That is reading. You close the chat. Can you do it again? If yes, you learned. If no, you read. Reading is useful. It is not the same as learning. Learning is what you can reproduce on a blank board. The blank board test is simple: close everything, take a blank page, reproduce the key step. If you can, the path is yours. If you cannot, the path was the model's. You followed it. You did not own it. The blank board test is the difference between following and owning.

The standard. In mathematics, reproducibility is the standard. A proof is yours if you can reproduce it. A proof is the model's if you cannot. The blank board test is the check: close everything, take a blank page, reproduce the proof. If you can, the path is in your hands. If you cannot, the path was in the model's hands and you were reading over its shoulder. Both are useful experiences. Only one is understanding. The student who adopts reproducibility as the standard will always know the difference. The student who does not will confuse following for owning for the rest of their education.

What the test reveals. The blank board test reveals exactly which steps you own and which you followed. You might own steps 1 through 3 and lose the path at step 4. That is useful data. Step 4 is the step you need to relearn. The model can help you relearn it. Then you take the blank board test again. If you can reproduce step 4 this time, you own it now. The test is not a grade. It is a diagnostic. It tells you where the path is yours and where it is not. Then you fix the gaps. The test is the map of your understanding.

Sitting big idea. If you cannot reproduce it on a blank board, you read it. You did not learn it. The blank board test is the check.

If they say “I understood it while the model was showing it.”: Understanding while reading is real. It is not the same as reproducing. The blank board test tells you which one you have. If you can reproduce it, you own it. If you cannot, you understood it while it was in front of you. Both are useful. Only one is learning.

If they say “I don't need to reproduce it. I understood it.”: Understanding while following is real. It is not the same as reproducing. The blank board test is the difference. If you can reproduce it, you own it. If you cannot, you understood it while it was in front of you. That is a different thing. Both are useful. Only one is mastery.

If they say “In the real world, I can just look it up.”: You can look up the steps. You cannot look up the understanding. The steps are on the internet. The path in your hands is not. The blank board test is how you know which one you have.

Integrity. Do not peek at the model's solution during the blank board test. If you peek, the test is invalid. The peek is the data: you could not reproduce it.

Dinner. What could we reproduce on a blank board today, and what could we not?

- Do not shame the child for failing the blank board test. The failure is data, not a grade.
- Keep it to one key step, not the whole problem. The point is the test, not the volume.

Adult sitting:

If you never reproduce a step in front of the child, they learn that math is something you read, not something you own. Show them the test.

Objective. The adult takes one blank board test on a household calculation in front of the child, names the result honestly, and models that reproducibility is the standard.

For the grown-up in the room. The adult models the blank board test. Do a household calculation with the model. Then close everything. Try to reproduce the key step on a blank page, in front of the child. If you can, say: 'I own this step.' If you cannot, say: 'I followed it. I need to learn it again.' The child who sees an adult take the blank board test honestly learns that reproducibility is the standard, not just a school rule. The child who sees an adult skip the test learns that math is something you read, not something you own. The modeling is the lesson.

The honest test. The adult who takes the blank board test in front of the child and passes teaches: reproducibility is achievable. The adult who takes it and fails, honestly, teaches: reproducibility is the standard, and not meeting it is data, not shame. Both are lessons. The adult who never takes the test teaches: math is something you read, not something you own. That is the wrong lesson. Take the test. Pass or fail. Name the result. The child learns that the standard exists and that honest adults hold themselves to it.

Sitting big idea. Take the blank board test in front of the child. Pass or fail. Name the result. The child learns that reproducibility is the standard.

If they say "I'll fail it in front of the kid.": Then you fail honestly. The child who sees an adult fail a blank board test and say 'I need to learn this again' learns more than from a perfect performance. They learn that the standard is real and that honest adults hold themselves to it. The failure is the lesson. The hiding is not.

Integrity. If you fail the blank board test, say so. Do not hide it. The honest failure teaches the standard. The hidden failure teaches nothing.

Dinner. What blank board test did we take this week, and what was the honest result?

- If you fail the blank board test, say so. Do not hide the failure. The honest failure is more instructive than a hidden success.

- Keep it to one key step. The point is the test, not the performance.

The Big Idea

Reproducibility is the standard. A proof you cannot reproduce is not yours. The blank board test is the map of what you own and what you followed. Reproducibility is not optional because the instrument is fluent.

Practice

1. One problem at the edge of skill. Path on paper. No tool. 2. Hunt a gap. Mark the missing step. 3. Tool as hostile reviewer of legal steps only. Reject a full solution. 4. Close. Blank-board one key step.

Reflect

If the chat closed, which step could you still walk — and which was a rental?

Chapter 5: Proof as a Social Act



Illustration for Chapter 5

A proof is a social act: you write a path so others can walk it. The model writes for itself. You write for the reader. The reader is the test.

Objective. The student can write a proof that a peer could follow, use the model as a test reader, and explain why proof is a social practice, not a private one.

For the grown-up in the room. This sitting teaches the student that proof is social. Mathematics is not a private act. It is a public path. You write so others can verify. The model writes for itself. The student writes for the reader. The reader is the test: a peer, a model used as a reader, or a teacher. The proof that only the writer can follow is a note. The proof a reader can walk is mathematics. The student who adopts this standard will write proofs that communicate, not just calculate.

Mathematics is social. Mathematics is not done in private. A proof is written for a community. The community verifies it. If the community cannot follow it, the proof is not accepted. This is not a school rule. It is how mathematics works. The model writes proofs that skip steps because it does not have a reader. It has a pattern. The student who writes for a reader is doing mathematics. The student who writes for themselves is keeping notes. The difference is the reader. Write for the reader. Use the model as a test reader. Use a peer as a real reader. The proof that survives both is a proof.

The model as a test reader. The model can be a test reader. You write the proof. You show the model. You ask: 'Could a peer follow this? Where would they get stuck?' The model finds the gaps. You fill them. This is different from asking the model to write the proof. The model as a reader tests your proof. The model as a writer replaces your proof. Use it as a reader. Never use it as a writer. The student who knows the difference is doing mathematics with a tool. The student who does not is outsourcing their thinking.

The path for the child. Write the household calculation as if the child were the reader. Show every step. Then show the child. Ask: 'Could you walk this?' If they can, the path is social: another person can verify it. If they cannot, the path is private: only you can follow it. Fill the gap. Make it social. The child who sees an adult write for a reader learns that math is a public path, not a private answer. That is the deeper lesson of this unit: mathematics is something we do together, not something we do alone with a machine.

Sitting big idea. Proof is social. Write for the reader. Use the model as a test reader, never as a writer. The proof that survives a reader is a proof.

If they say "I don't need to write for others. I understand it.": You understand it for yourself. That is a note. A proof is for the reader. If you cannot write it so another person can follow, you have a private understanding, not a public proof. Mathematics is public. Write for the reader.

If they say “The model can just write the proof.”: It can. That proof is the model's, not yours. It is written for the model's pattern, not for your reader. Use the model as a test reader. Write the proof yourself. The proof that is yours is the one you wrote for a reader.

If they say “The child can't follow my math.”: Then write simpler steps. The path is for the reader. If the reader is the child, write for the child. If they cannot follow, the steps are too high. Lower them. The point is not the complexity. It is the walkability.

Integrity. Cite the model as a test reader if it found gaps you fixed. The proof is yours. The review is the model's. The citation is honest.

Dinner. What proof did we write for a reader this week, and did it survive the test?

- Academic integrity: the proof is yours if you wrote it. If the model wrote it and you copied, it is the model's proof, not yours.

- The model as a test reader is different from the model as a writer. The student must use it to test, not to write.

Young sitting:

A proof is not for you. It is for the person who needs to walk it after you. Write it so they can.

Objective. The student can write a simple proof in steps that another person could follow without help, and explain why a proof is for the reader, not the writer.

For the grown-up in the room. A proof is a social act. It is not for you. It is for the person who reads it. If only you can follow it, it is a note, not a proof. A proof is written so that someone else can walk the path. The model writes proofs for itself. It jumps. It skips. It assumes. The student needs to learn to write for the reader. The reader is the test: could a friend walk your steps? If yes, it is a proof. If no, it is a note.

The reader is the test. A proof is a path. The path is for the person who walks it after you. If you write the proof for yourself, you will skip steps that are obvious to you. The reader will fall in the gap. If you write for the reader, you show every step, even the ones that feel obvious. The test is simple: could a friend who has not seen this problem walk your steps and arrive at your answer? If yes, it is a proof. If no, it is a note to yourself. Both are useful. Only one is a proof.

The model writes for itself. The model writes proofs that skip steps because the steps are obvious to the model's pattern. They are not obvious to you. The model's proof is a note to itself, not a proof for you. When you read it and cannot follow, the gap is not your failure. It

is the model's choice to write for itself, not for you. The skill is to read the model's proof and fill the gaps it left for itself. Then write your own proof, for your reader, with every step shown.

Mathematics is social. Mathematics is not done in private. A proof is written for a community. The community verifies it. If the community cannot follow it, the proof is not accepted. This is not a school rule. It is how mathematics works. The model writes proofs that skip steps because it does not have a reader. It has a pattern. The student who writes for a reader is doing mathematics. The student who writes for themselves is keeping notes. The difference is the reader. Write for the reader. Use the model as a test reader. Use a peer as a real reader. The proof that survives both is a proof.

The model as a test reader. The model can be a test reader. You write the proof. You show the model. You ask: 'Could a peer follow this? Where would they get stuck?' The model finds the gaps. You fill them. This is different from asking the model to write the proof. The model as a reader tests your proof. The model as a writer replaces your proof. Use it as a reader. Never use it as a writer. The student who knows the difference is doing mathematics with a tool. The student who does not is outsourcing their thinking.

Sitting big idea. A proof is for the reader, not the writer. Write it so a friend could walk it. The model writes for itself. You write for others.

If they say "It's obvious to me.": It is obvious to you. Is it obvious to the reader? The proof is for the reader. If the reader cannot follow, the proof is a note, not a proof. Fill the step.

If they say "I don't need to write for others. I understand it.": You understand it for yourself. That is a note. A proof is for the reader. If you cannot write it so another person can follow, you have a private understanding, not a public proof. Mathematics is public. Write for the reader.

If they say "The model can just write the proof.": It can. That proof is the model's, not yours. It is written for the model's pattern, not for your reader. Use the model as a test reader. Write the proof yourself. The proof that is yours is the one you wrote for a reader.

Integrity. Do not present the model's proof as your own. If the model wrote the steps and you filled the gaps, cite both. The proof is yours if you wrote it for the reader.

Dinner. What proof did we write for a reader this week, and could they walk it?

- Keep it to simple proofs: arithmetic, a basic geometric argument.
- Do not grade the child's proof style. Grade whether another person could follow it.

Adult sitting:

If you only show answers, the child learns that math is private. If you show paths for others, they learn that math is social. That is the deeper lesson.

Objective. The adult models writing a path for a reader on one real household calculation, shows it to the child as if the child were the reader, and names that the path is for the reader.

For the grown-up in the room. The adult models the social act of proof. Do a household calculation. Write the steps as if the child were the reader. Show the child. Ask: 'Could you walk these steps?' If they can, the path is social. If they cannot, the path is private. The adult who writes for the reader teaches the child that math is a social act: you show your path so others can verify. The adult who only shows answers teaches the child that math is private: you have the answer and the path is yours alone. The social path is the lesson.

The path for the child. Write the household calculation as if the child were the reader. Show every step. Then show the child. Ask: 'Could you walk this?' If they can, the path is social: another person can verify it. If they cannot, the path is private: only you can follow it. Fill the gap. Make it social. The child who sees an adult write for a reader learns that math is a public path, not a private answer. That is the deeper lesson of this unit: mathematics is something we do together, not something we do alone with a machine.

Sitting big idea. Write the path for the child. If they can walk it, the path is social. That is what mathematics is: a path others can verify.

If they say "The child can't follow my math.": Then write simpler steps. The path is for the reader. If the reader is the child, write for the child. If they cannot follow, the steps are too high. Lower them. The point is not the complexity. It is the walkability.

Integrity. If you used the model for the answer, say so. The path is yours if you wrote the steps. The answer is the model's if you asked it. The child can tell.

Dinner. What path did we write for the child to walk this week, and could they walk it?

- Do not lecture. Write the path. Show it. Ask the child if they can walk it. That is the lesson.
- If the child cannot walk it, that is data. Fill the gap. The gap is the lesson.

The Big Idea

Proof is social. Write for the reader. Use the model as a test reader, never as a writer. The proof that survives a reader is a proof. Reproducibility is not optional because the instrument is fluent.

Practice

1. One problem at the edge of skill. Path on paper. No tool. 2. Hunt a gap. Mark the missing step. 3. Tool as hostile reviewer of legal steps only. Reject a full solution. 4. Close. Blank-board one key step.

Reflect

If the chat closed, which step could you still walk — and which was a rental?

Chapter 6: The Full Path



Illustration for Chapter 6

Walk, write for a reader, check with the model, reproduce on a blank board, report. One full mathematical path, end to end. That is the capstone.

Objective. The student can run the full proof cycle on a real mathematical problem, produce a readable proof that survives a model's hostile review and a blank board test, and write a one-paragraph report.

For the grown-up in the room. This is the emerging capstone. The student runs the full cycle: solve, write for a reader, submit to the model's hostile review, reproduce on a blank board, report. The output is one proof that survived three tests: the reader test, the model's review, and the blank board. The proof is the capstone artifact. It is the proof that the student can show a path others can walk in the age of the machine that gives the answer.

The three-test proof. The capstone proof passes three tests. The reader test: a peer can walk it. The model's hostile review: the model found the weak steps and the student fixed them. The blank board test: the student can reproduce it without notes. A proof that passes all three is a proof. A proof that passes one or two is a draft. The student who runs the cycle and passes all three has produced something the model cannot: a path that is owned, verified, and reproducible. That is mathematics in the age of fluency.

One cycle at the table. The adult runs the cycle on a real calculation. Walk the steps. Write them for the child as the reader. Check with the model. Reproduce on a blank board. Report. The child sees: math is a path, the model checks, the blank board tests, and the path is for the reader. That is the whole unit, modeled in one cycle, at the table, on something real. The child will not remember the vocabulary. They will remember the Tuesday you walked a path, checked it, reproduced it, and said: 'That is the path. The answer is just where the path ends.'

Sitting big idea. The capstone proof passes three tests: the reader, the model's review, and the blank board. A proof that passes all three is a proof the model cannot replace.

If they say "This is too much work for one proof.": The first time, it is. The capstone is the investment. The second proof will be faster. The third will be a habit. The capstone is the proof that the skill exists. After that, the skill is yours.

If they say "The model can write a better proof.": It can write a smoother proof. It cannot write a proof that passes your blank board test. It cannot write a proof that a peer walks. It can write a proof that a reader follows, but the path is the model's, not yours. The capstone is the proof that is yours.

If they say "The child won't follow all of it.": They do not need to follow all of it. They need to see it. The seeing teaches: math is a cycle, not an answer. The understanding comes later. The habit comes from seeing it once.

If they say “I don't have time for the full cycle at home.”: The first time takes fifteen minutes. The second time takes eight. The third time is a habit. One cycle, once, out loud, is the investment. The return is a child who shows paths instead of collecting answers.

Integrity. The report cites the model as a reviewer, the peer as a reader, and the blank board as the reproduction test. The proof is the student's. The citations are honest.

Dinner. What proof did we build this week, and did it pass all three tests?

- The proof must survive all three tests. If it fails one, the student fixes it and retries. The capstone is the proof that passed.

- Academic integrity: the proof is the student's. The model is cited as a reviewer. The blank board is the student's own reproduction.

Young sitting:

You now know: the path is the proof, the model checks, the blank board tests, and the path is for others. Show one full path.

Objective. The student can walk one full mathematical path: solve, write steps for a reader, check with the model, reproduce on a blank board, and report.

For the grown-up in the room. This is the capstone. The student runs the full cycle: solve a problem, write the steps for a reader, check with the model, reproduce on a blank board, report. The cycle ties together every skill from the unit. The output is one path, shown end to end, that a reader could walk and the student could reproduce. That is the capstone artifact.

The whole unit in one path. Walk a path. Write it for a reader. Check it with the model. Reproduce it on a blank board. Report. That is the whole unit in one cycle. The path is the proof. The model is the checker. The blank board is the test. The reader is the social. The report is the honesty. The child who can run this cycle once has learned the unit. The child who runs it again has a habit. The child who runs it for life has a skill that the model cannot replace: the ability to show a path others can walk.

The three-test proof. The capstone proof passes three tests. The reader test: a peer can walk it. The model's hostile review: the model found the weak steps and the student fixed them. The blank board test: the student can reproduce it without notes. A proof that passes all three is a proof. A proof that passes one or two is a draft. The student who runs the cycle and passes all three has produced something the model cannot: a path that is owned, verified, and reproducible. That is mathematics in the age of fluency.

Sitting big idea. Walk, write for a reader, check, reproduce, report. The whole unit in one path. That is the capstone.

If they say “That’s a lot for one problem.”: The first time, it is. The second time, it is faster. The third time, it is a habit. The cycle is an investment. The return is a path you own, that a reader can walk, and that the model checked. That is worth twenty minutes.

If they say “This is too much work for one proof.”: The first time, it is. The capstone is the investment. The second proof will be faster. The third will be a habit. The capstone is the proof that the skill exists. After that, the skill is yours.

If they say “The model can write a better proof.”: It can write a smoother proof. It cannot write a proof that passes your blank board test. It cannot write a proof that a peer walks. It can write a proof that a reader follows, but the path is the model’s, not yours. The capstone is the proof that is yours.

Integrity. The report includes the path, the model’s check, and the blank board result. Nothing is erased. The full cycle is the honest report.

Dinner. What full path did we show this week, and what did the cycle reveal?

- The student must run every step. Skipping a step is skipping the capstone.
- Celebrate the cycle, not the correctness. A wrong answer with a complete path is more valuable than a right answer with no path.

Adult sitting:

Run the full cycle once, at the table, on something real. The child who sees the cycle will run it for life. That is the capstone.

Objective. The adult runs the full proof cycle on one real household calculation in front of the child, reports the result out loud, and models that the path is the skill.

For the grown-up in the room. This is the adult capstone. The adult runs the full cycle: solve, write for a reader (the child), check with the model, reproduce on a blank board, report. Out loud. At the table. On one real calculation. The child sees the full cycle once. That is the investment. The child who sees an adult walk a path, check it, reproduce it, and report it honestly will internalize the cycle. The child who never sees it will use the model as a walker for life. One cycle. Out loud. That is the capstone.

One cycle at the table. The adult runs the cycle on a real calculation. Walk the steps. Write them for the child as the reader. Check with the model. Reproduce on a blank board. Report. The child sees: math is a path, the model checks, the blank board tests, and the path

is for the reader. That is the whole unit, modeled in one cycle, at the table, on something real. The child will not remember the vocabulary. They will remember the Tuesday you walked a path, checked it, reproduced it, and said: 'That is the path. The answer is just where the path ends.'

Sitting big idea. Run the full cycle once, at the table. The child who sees it will run it for life. The answer is where the path ends. The path is the skill.

If they say “The child won't follow all of it.”: They do not need to follow all of it. They need to see it. The seeing teaches: math is a cycle, not an answer. The understanding comes later. The habit comes from seeing it once.

If they say “I don't have time for the full cycle at home.”: The first time takes fifteen minutes. The second time takes eight. The third time is a habit. One cycle, once, out loud, is the investment. The return is a child who shows paths instead of collecting answers.

Integrity. If a test fails, say so. The honest cycle is the lesson. The cycle that hides a failure is a performance, not a model.

Dinner. What full cycle did we run at the table this week, and what did the child see?

- Do not make this a lecture. Run the cycle. Talk through the steps. The child watches.
- If a test fails, say so. The honest failure is the lesson. The hidden failure is not.

The Big Idea

The capstone proof passes three tests: the reader, the model's review, and the blank board. A proof that passes all three is a proof the model cannot replace. Reproducibility is not optional because the instrument is fluent.

Practice

1. One problem at the edge of skill. Path on paper. No tool. 2. Hunt a gap. Mark the missing step. 3. Tool as hostile reviewer of legal steps only. Reject a full solution. 4. Close. Blank-board one key step.

Research Prompt

Read Pólya, *How to Solve It*, the first page of the “How to Solve It” list (unknown / data / condition). Write one page mapping those three questions onto one proof you actually walked this week. Do not ask a model to write the page. Cite Pólya.

Debates You Should Be Able to Hold

“In the real world we use tools.” Someone still has to know when the tool is insane. Stand: checker, not walker.

“Proofs are just for contests.” A proof is for other people — lab, code, money. Stand: social act.

“The camera is showing work.” A screenshot is a prize. Stand: blank board.

“Everyone will use it anyway.” Then the ones who can still walk will be trusted. Be those.

For the Grown-Up Reader

High band. Young+ only. Pólya 1945. Little 404. *Number in the Age of AI* is a different pack.

Slugs: `what-is-proof`, `missing-step`, `checker-not-walker`, `blank-board`, `proof-social`, `show-path-capstone`.

About WisdomForge

This is the high band of *Show the Path*. Payment pulled.

Sources

Primary. George Pólya, *How to Solve It* (Princeton, 1945). WisdomForge sittings, *Show the Path*.

Secondary. Feynman, “Cargo Cult Science,” 1974 — a boxed answer can imitate a path.